

Experiment 1

Installation of Hypervisors and Initiation of VMs

VirtualBox + Ubuntu Virtual Machine

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Course & Semester	Cloud Computing Lab (CSE12045) · 7th Semester
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1. Objective

Install and configure an open-source hypervisor, create and instantiate an Ubuntu Virtual Machine using an ISO image, configure the virtual hardware, boot the guest operating system, verify successful execution, and install the required build tools and Linux kernel headers.

2. Requirements

Oracle VirtualBox and a suitable Ubuntu ISO image.

3. Procedure

Step 1 — Create a New Virtual Machine

Open Oracle VirtualBox and select New. Enter the VM name and select the downloaded Ubuntu ISO image. The captured configuration shows the VM name as ubuntu-server and the Ubuntu ISO image selected.

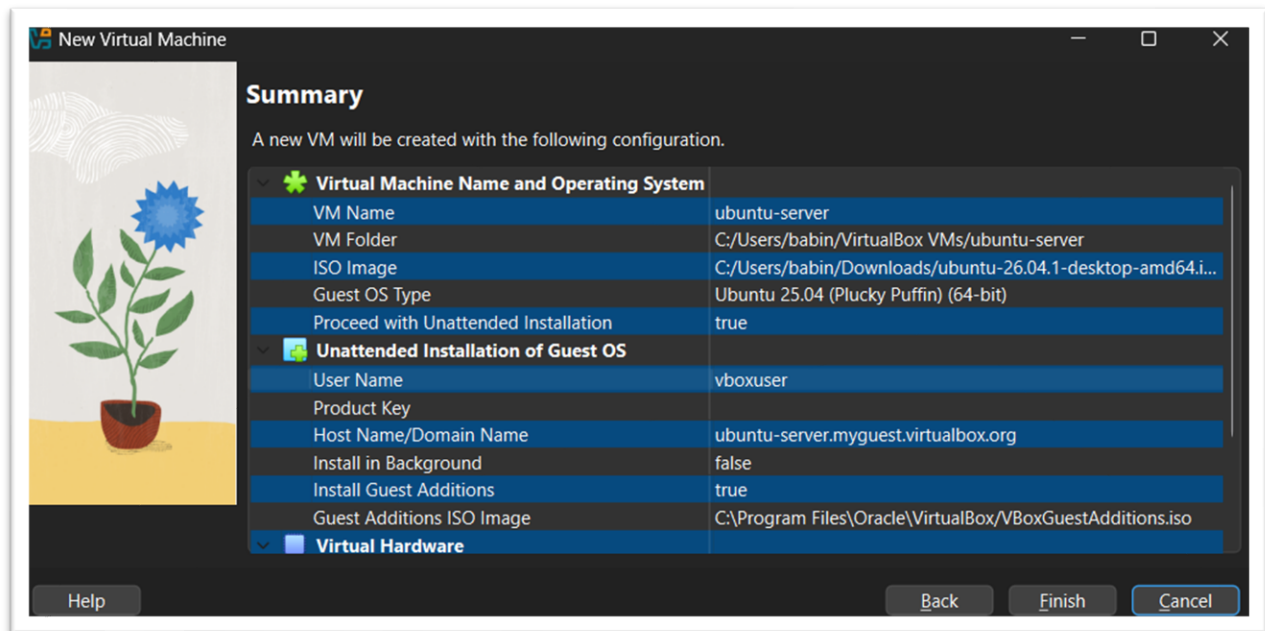


Figure 1 — VirtualBox VM creation summary showing VM name, ISO image, guest OS and unattended installation settings.

Step 2 — Configure Unattended Guest OS Installation

Enter the guest username and password, set the host name, and keep Guest Additions selected as required by the VirtualBox setup.

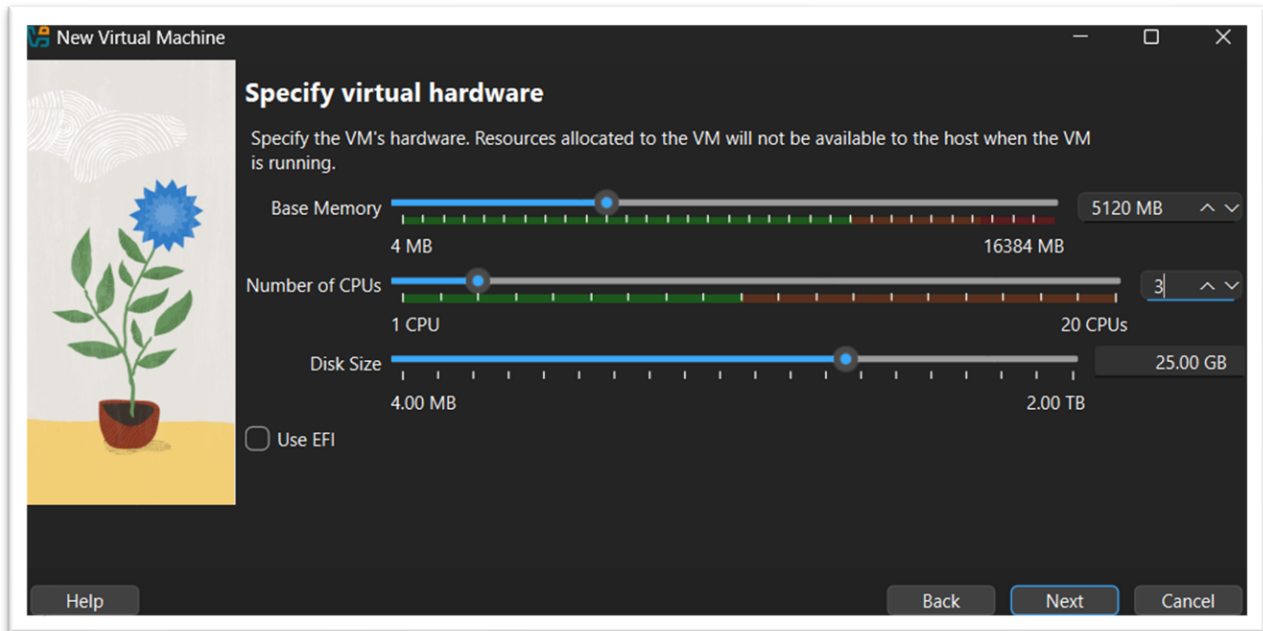


Figure 2 — Unattended guest OS installation settings.

Step 3 — Configure Virtual Hardware

Allocate suitable memory, CPUs and virtual disk space. The captured configuration uses 5120 MB RAM, 3 CPUs and a 25.00 GB virtual disk.

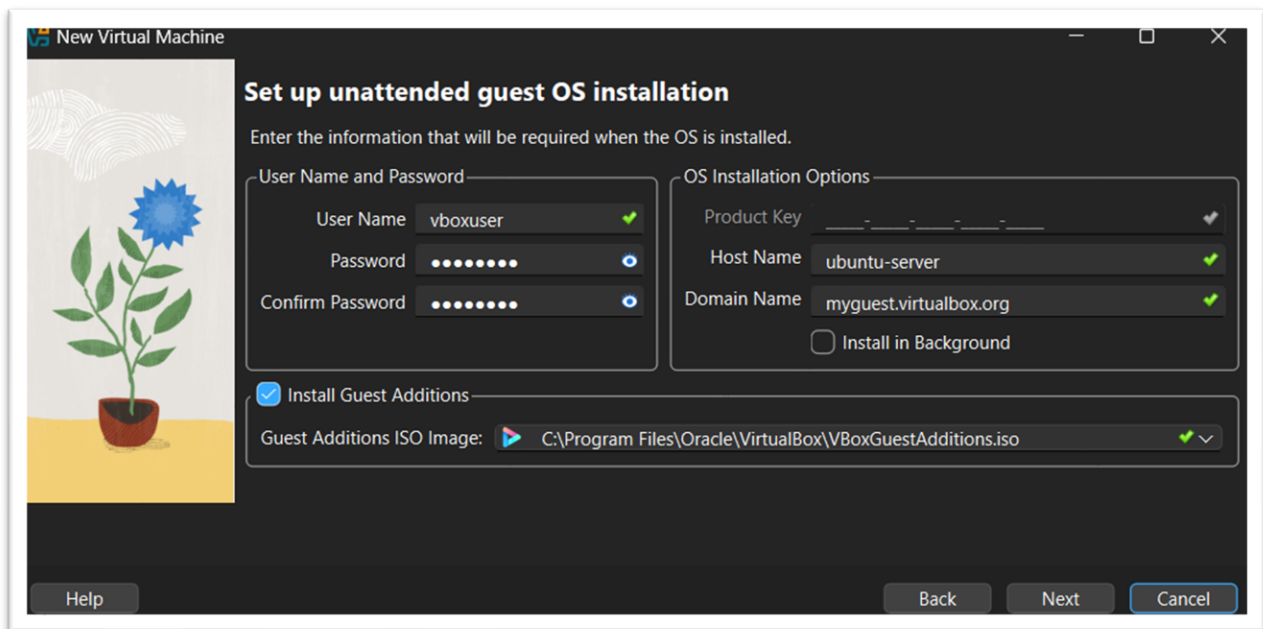


Figure 3 — Virtual hardware configuration.

Step 4 — Review and Create the VM

Review the complete configuration and click Finish to create the virtual machine.

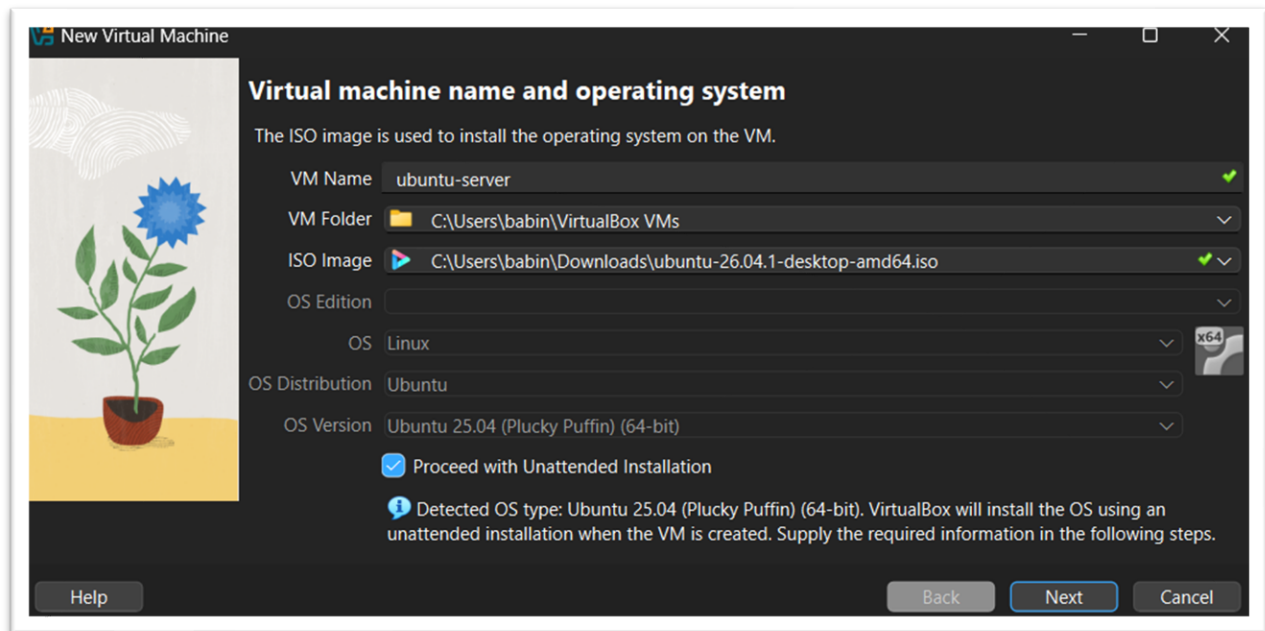


Figure 4 — Final VirtualBox configuration summary before VM creation.

Step 5 — Boot Ubuntu

Start the newly created VM and wait for Ubuntu to boot successfully.

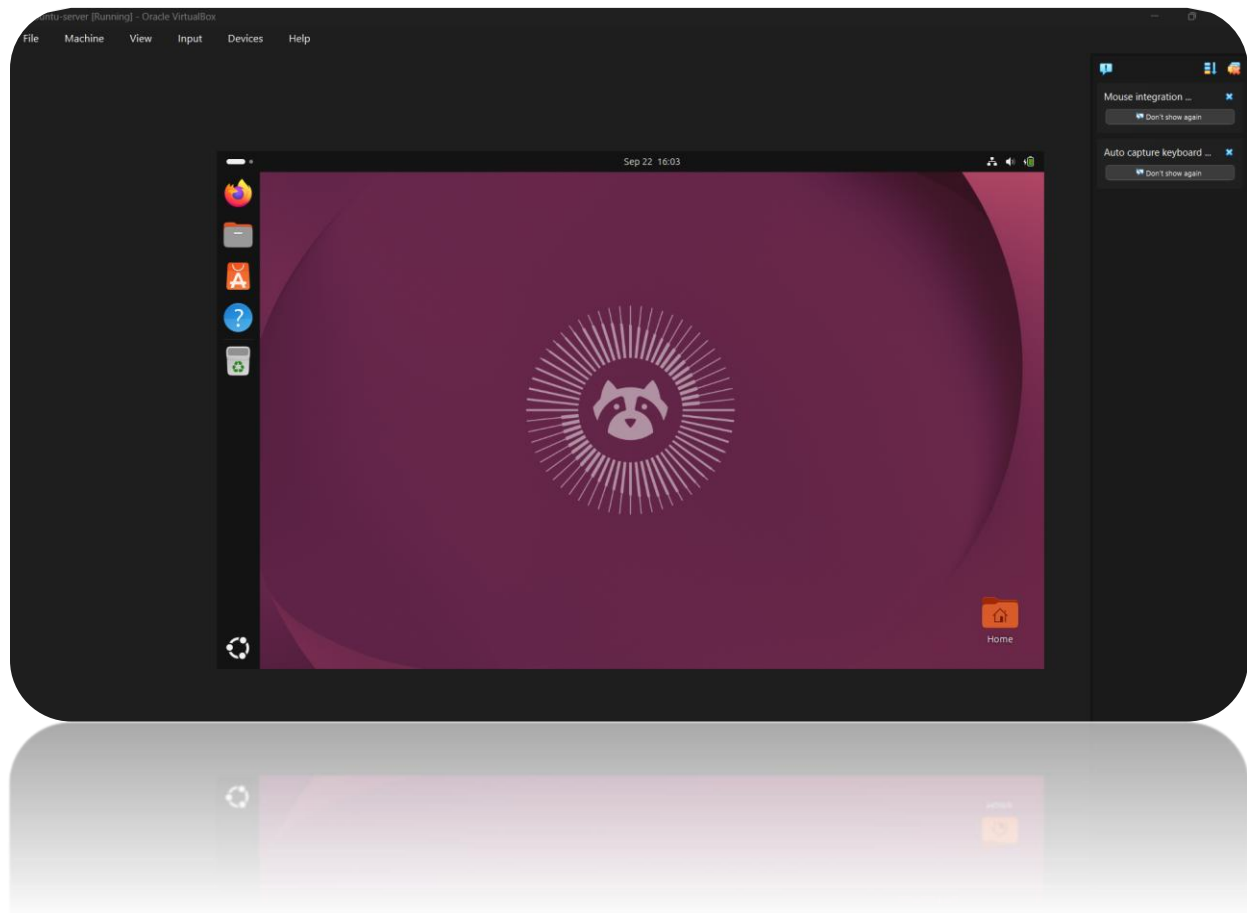


Figure 5 — Ubuntu virtual machine successfully booted inside VirtualBox.

Step 6 — Update Package Information

Open the Ubuntu terminal and execute the following command:

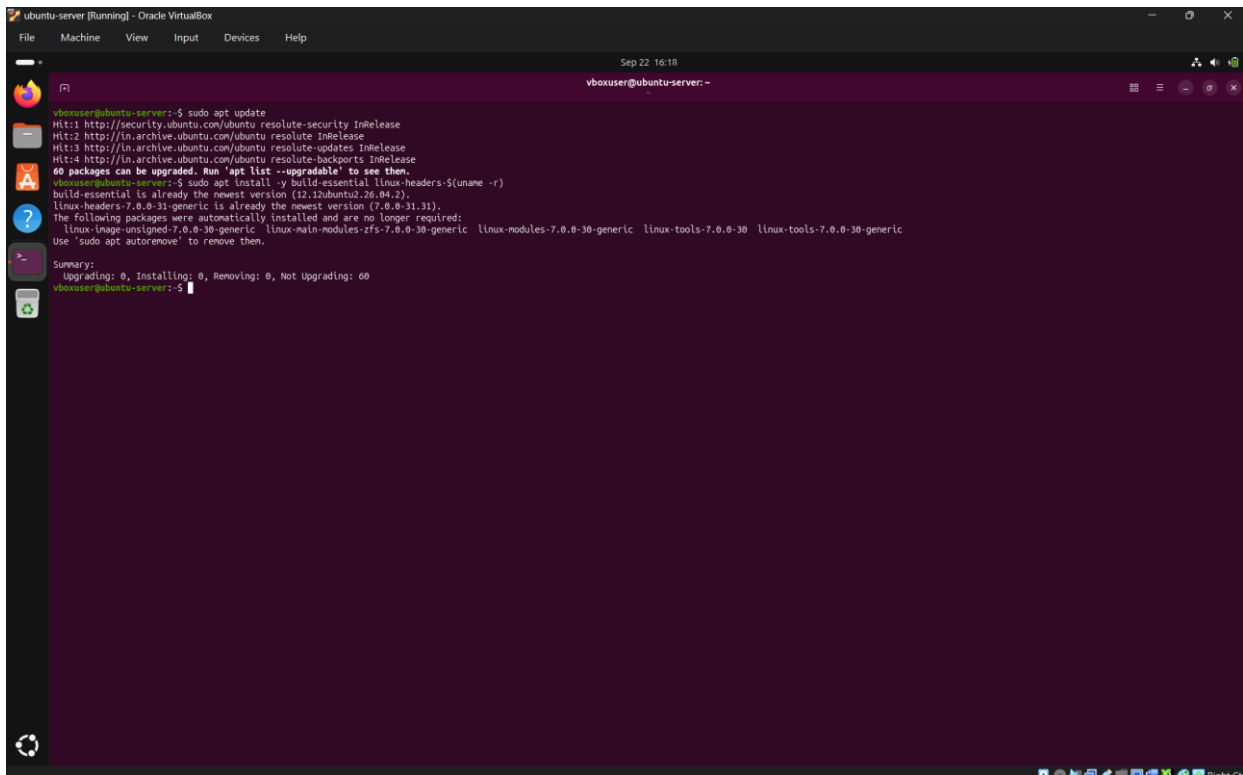
```
sudo apt update
```

The captured terminal output shows successful contact with the configured Ubuntu repositories.

Step 7 — Install Required Build Tools and Linux Kernel Headers

```
sudo apt install -y build-essential linux-headers-$(uname -r)
```

The captured terminal output confirms that build-essential and the Linux kernel headers were already at the newest available versions on the VM, so no additional packages needed to be installed at that point.



```
ubuntu-server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 22 16:18
vboxuser@ubuntu-server:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
60 packages can be upgraded. Run 'apt list --upgradable' to see them.
vboxuser@ubuntu-server:~$ sudo apt install -y build-essential linux-headers-$(uname -r)
build-essential is already the newest version (12.1ubuntu2.6.04.2).
linux-headers-7.0.0-31-generic is already the newest version (7.0.0-31.31).
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.0.0-30-generic linux-main-modules-zfs-7.0.0-30-generic linux-modules-7.0.0-30-generic linux-tools-7.0.0-30 linux-tools-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 60
vboxuser@ubuntu-server:~$
```

Figure 6 — Ubuntu terminal showing package update and verification/installation of build-essential and Linux kernel headers.

4. Result

The Ubuntu virtual machine was created and configured successfully in Oracle VirtualBox. The VM booted successfully, package information was updated, and the required build tools and Linux kernel headers were verified.

5. Conclusion

A functional Ubuntu virtual machine environment was prepared successfully. This VM provides the required environment for the subsequent Cloud Computing Lab networking experiments.